

Water and Environmental Impact of AI usage

Posted by u/Professionalcy9543 • • 0 points • 68 comments

Hey ya'll. I've lurked on this sub for a few months now and I keep seeing arguments about AI's environmental and water impacts. I'm anti-AI by principle but I think I can bring some expertise to the table, for both sides.

While I am some random person on the internet and you have little reason to actually trust that my credentials are true, I will list them regardless as they have given me salient education and training for this subject. I've worked with the power industry for 13 years. I have worked in the United States, UAE, Guyana and most recently, the UK. I've worked mostly with the oil and the nuclear industries.

With that out of the way, the first common stumbling block for this is water usage. A lot of people on the anti-AI side use hyperbole to describe the amount of water used to create single AI images. On the pro side, I see a lot of minimization of water consumption. I'm going to stick with a source I trust, the OECD.

<https://oecd.ai/en/wonk/how-much-water-does-ai-consume>

While creating individual images only uses a small amount of water in the grand scheme of things, maintaining the power and server infrastructure to allow you to create images and text on demand takes a massive amount of water per day. Now. This is a problem within the power industry as well. Power plants for the most part use a lot of water daily. So, with AI datacenters being put into grids, it creates a feedback loop, of these centers needing water for coolant, and the powerplants that feed them needing more water for the higher loads they experience. But wait! It's not all bad. This water doesn't all just go away. However, there is a important distinction made in the link above. Consumption vs Withdrawal

Water consumption is when you remove water from a river or groundwater deposit or some such source, use it, and let it percolate back into the environment. It's temporary, and with proper treatment, not a big deal. You just cool it off and let it go back in once it's a safe temp. Sometimes you might need to treat it a little based on the scale, but for the most part, no biggie. This is what happens with cooling ponds and towers in power plants.

Water withdrawal is when it doesn't go back into the local supply. This is when water that is cooling or otherwise stored in a way that it can evaporate evaporates. This water doesn't

become unusable, but it usually doesn't remain in one place. This effect scales. The more water you need, the more you end up losing in treatment. This is a big problem. With the scaling and higher and higher server and energy needs in the AI industry, the amount of water being withdrawn from municipalities that AI farms exist in increases exponentially.

[<https://news.lenovo.com/data-centers-worlds-ai-generators-water-usage/>]

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[https://apnews.com/article/chatgpt-gpt4-iowa-ai-water-consumption-microsoft-f551fde98083d17a7e8d904f8be822c4?utm_source=copy&utm_medium=share]

(https://apnews.com/article/chatgpt-gpt4-iowa-ai-water-consumption-microsoft-f551fde98083d17a7e8d904f8be822c4?utm_source=copy&utm_medium=share)

This poses a looming issue. Without a concrete plan to reduce or mitigate water losses, the scale of AI can very quickly destroy water supplies. While the water isn't being destroyed per se, it is being removed from the community that these exist in. This is a serious threat. These aquifers replenish but very very slowly, way slower than the rate they are being tapped in.

This also doesn't have an individual scale. You might not spend a lot of water per generated image, but to maintain that infrastructure to allow you to do so is incredibly costly. It's the tragedy of the commons.

[https://en.wikipedia.org/wiki/Tragedy_of_the_commons]

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You may have a small effect, but others acting the same way can quickly create a crisis. For more information, this is an excellent article by the ASCE:

<https://www.asce.org/publications-and-news/civil-engineering-source/civil-engineering-magazine/issues/magazine-issue/article/2024/03/engineers-often-need-a-lot-of-water-to-keep-data-centers-cool>

My other main concern is E-Waste. Server farms don't last forever. Components fail or need to be upgraded and that poses a huge issue. Most computer technology is difficult to recycle and contains a lot of dangerous elements and chemicals. A study by Nature shows that this could result in over 1.3 million tons of E-Waste being created in the next 5 years alone, solely by AI:

[<https://www.nature.com/articles/s43588-024-00712-6>]

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This is a problem that neither side is talking about. Improper disposal of E-Waste can kill entire regions. It can devastate ecosystems and towns if not properly taken care of. Modern facilities simply do not have the capacity to take this load on in the next several years. There needs to be investment by not just the government but by AI companies to prepare for this.

You all don't have to believe me. I've cited my sources. TLDR, if there is not significant investment done in water retention and E-Waste disposal, we could end up seeing towns run out of water, and counties polluted to the point that people can't live there anymore. This isn't unique to AI, but it's definitely being accelerated by it, far past our current ability to deal with it.

Comments

u/xiaojunprivateequity • 1 points • 2025-12-23 21:12 UTC

all global data centers will use 250000-500000 acrefeet of water in 2025. in 2005, golfcourses used 2.1 million acrefeet of water per year (thus, more now). average family household uses from .25-1 acrefeet of water a year. bottled water uses 50 thousand acrefeet a year.

u/farrahlynnflowers • 1 points • 2025-12-12 05:26 UTC

This should be cross promoted